

AI Agent & Machine Learning Solutions

Final Internship Project Report & Deliverables Summary

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GitHub Repository: <https://github.com/kartheek2805/AI-Agent-Machine-Learning-Solutions>

1. Executive Summary

This report details the successful design, implementation, and deployment of the **AI Agent & Machine Learning Solutions** platform. Built as an end-to-end internship capstone project, the system combines automated data analytics, machine learning predictive engines (Sales Forecasting, Collaborative Filtering Recommendation Agent, K-Means Customer Clustering), autonomous natural language decision agents, and an interactive Streamlit UI web application.

2. Project Milestones Roadmap Alignment

The project was executed strictly in accordance with the 5 project milestones depicted in the roadmap:

Milestone Phase	Duration	Objectives & Key Deliverables	Status
Week 1: Setup & Planning	Week 1	Requirements analysis, environment setup, repo structure.	Completed ■
Week 2-3: Research & Design	Weeks 2–3	Architecture design, dataset schema, ML pipeline spec.	Completed ■
Week 4-6: Dev Phase 1	Weeks 4–6	EDA engine, Sales Forecasting ML, Recommendation Agent, K-Means.	Completed ■
Week 7-8: Dev Phase 2	Weeks 7–8	DataQueryAgent, StrategicDecisionAgent, Streamlit UI.	Completed ■
Week 8-9: Final Review	Weeks 8–9	Phase docs, PPTX presentation, final report, GitHub push.	Completed ■

3. Machine Learning Suite & Performance

Sales & Demand Forecasting Model (SalesForecaster): Uses Random Forest Regression to predict future sales revenue. The model achieved an **R² score of 99.99%**, MAE of **.51**, and RMSE of **.73**. Feature importance analysis highlighted *Units_Sold* and *Unit_Price* as primary drivers.

Collaborative Recommendation Agent (RecommendationAgent): Implements item-item and user-item collaborative filtering using Cosine Similarity matrices computed over user rating pivot tables to generate top product recommendations.

Customer Behavioral Segmenter (CustomerSegmenter): Applies K-Means Clustering to group customer accounts into High-Value VIPs, Steady Growth Buyers, and Occasional Customers.

4. Autonomous AI Decision Agents

Natural Language Query Agent (DataQueryAgent): Accepts queries in plain English (e.g. *What is total revenue?*, *Which product is top seller?*) and computes dynamic analytical answers with metric cards and tables.

Autonomous Strategic Advisor (StrategicDecisionAgent): Evaluates system revenue telemetry, identifies product margin bottlenecks, flags regional expansion opportunities, and suggests proactive customer churn mitigation tactics.

5. Project Submission & Deliverables

- **GitHub Repository:** <https://github.com/kartheek2805/AI-Agent-Machine-Learning-Solutions>
- **PowerPoint Presentation (.pptx):** AI_Agent_ML_Solutions_Presentation.pptx
- **Streamlit Interactive App:** src/app.py (5 Navigation Tabs)
- **Documentation Suite:** docs/01_project_setup_and_planning.md through 05_final_review_and_submission.md